

Chapter 1

Title

Retail Sales Performance Dashboard

Subtitle

A Business Intelligence Solution using Microsoft Power BI

Prepared By

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Role

Data Analyst Portfolio Project

Tools Used

- **Microsoft Power BI**
- **Power Query**
- **DAX**
- **Microsoft Excel**

Submission Date

July 2026

Chapter 2

Certificate

I hereby declare that the project titled "**Retail Sales Performance Dashboard**" is my original work completed as part of my Data Analytics portfolio. The analysis, dashboard development, and documentation presented in this report were created using Microsoft Power BI, Power Query, DAX, and Microsoft Excel for educational and portfolio purposes.

Prepared By

Vikas Gaikwad

Chapter 3

Acknowledgement

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this project.

This project provided me with practical experience in business intelligence, data cleaning, data modelling, DAX, dashboard development, and business analysis using Microsoft Power BI.

I would also like to acknowledge the creators of the Kaggle Retail Sales Dataset, which served as the foundation for this analysis.

Finally, I appreciate the learning resources provided by Microsoft Learn, the Power BI community, and other educational platforms that helped me strengthen my analytical skills.

Chapter 4

Executive Summary

Organizations generate thousands of sales transactions every day, making it difficult to monitor business performance using traditional reports. Interactive dashboards help decision-makers analyze large volumes of data quickly and identify important business trends.

This project focuses on analyzing retail sales data from January 2024 to September 2024 using Microsoft Power BI.

The project involved:

- Cleaning and validating the dataset
- Creating an optimized data model
- Developing DAX measures
- Designing an interactive dashboard
- Identifying business insights
- Providing actionable recommendations

The final dashboard enables stakeholders to monitor key performance indicators (KPIs), evaluate sales trends, analyze customer demographics, compare regional performance, and support data-driven decision-making.

Chapter 5

Introduction

Business Intelligence (BI) enables organizations to convert raw data into meaningful insights that support strategic and operational decision-making. Modern BI tools such as Microsoft Power BI allow businesses to analyze large datasets through interactive dashboards and visualizations.

Retail organizations generate significant volumes of transactional data every day. Without a centralized reporting solution, identifying sales trends, monitoring business performance, and understanding customer behavior can be challenging.

The objective of this project is to demonstrate how Power BI can transform raw retail sales data into an interactive dashboard that provides valuable business insights.

Chapter 6

Company Background

Daily Discover Retail Pvt. Ltd. is a fictional retail company that sells products across multiple regions through physical stores and online channels.

The company offers products from several categories, including:

- Books
- Movies
- Electronics
- Toys
- Clothing
- Appliances
- Personal Care
- Home & Kitchen

The management team requires a centralized dashboard to monitor sales performance, evaluate customer behavior, analyze regional trends, and support business decisions.

Chapter 7

Business Problem Statement

The organization faces several reporting challenges:

- Sales information is scattered across transactional data.
- Managers spend significant time preparing reports manually.
- There is limited visibility into sales performance across regions and product categories.
- Customer purchasing behavior is difficult to analyze.
- Business decisions are delayed because insights are not available in real time.

To address these challenges, an interactive Power BI dashboard was developed.

Chapter 8

Business Objectives

The primary objectives of this project are:

1. Monitor overall sales performance.
2. Track key business KPIs.
3. Analyse sales trends over time.
4. Identify top-performing product categories.
5. Evaluate regional sales performance.
6. Understand customer demographics.
7. Analyze sales representative performance.
8. Support business decision-making through interactive dashboards.

Chapter 9

Dataset Information

Source

Kaggle – Retail Sales Dataset

Analysis Period

January 2024 – September 2024

Attribute	Value
Original Records	100,000
Final Records	90,000
Removed Records	10,000
Missing Values	Removed before import

Main Columns

- Order Date
- Product Category
- Sales Amount
- Discount
- Customer Age
- Customer Gender
- Sales Region
- Sales Representative

Data Collection

Dataset Source

The dataset used for this project was obtained from Kaggle, a popular platform that provides real-world datasets for analytics and machine learning projects.

The dataset contains retail sales transactions, including customer details, product information, sales amount, discount, region, and sales representative information.

The data covers the period from January 2024 to September 2024.

Dataset Characteristics

Attribute	Value
Source	Kaggle
File Format	CSV
Original Records	100,000
Final Records Used	90,000
Time Period	January 2024 – September 2024
Tool Used	Microsoft Excel & Power BI

Data Cleaning

High-quality data is essential for producing reliable dashboards and accurate business insights. Before importing the data into Power BI, the dataset was reviewed and cleaned to remove incomplete records and improve data quality.

Cleaning Process

The original dataset contained 100,000 records.

During data inspection in Microsoft Excel, it was observed that 10,000 records contained missing values in four critical columns:

- Sales Amount
- Discount
- Customer Gender
- Customer Age

Since these four columns were missing together in the same records, the affected rows could not contribute meaningful information to the **analysis**.

These incomplete records were removed before importing the dataset into Power BI.

After cleaning, the dataset contained 90,000 records, which became the final dataset used for analysis.

Removing incomplete records helped:

- Improve overall data quality.
 - Prevent inaccurate KPI calculations.
 - Avoid misleading visualizations.
 - Ensure reliable business insights.
 - Maintain consistency throughout the dashboard.
-

Dataset Summary

Description	Count
-------------	-------

Original Records	100,000
------------------	---------

Removed Records	10,000
-----------------	--------

Final Records	90,000
---------------	--------

Duplicate Records	Removed if found
-------------------	------------------

Missing Values	Eliminated before import
----------------	--------------------------

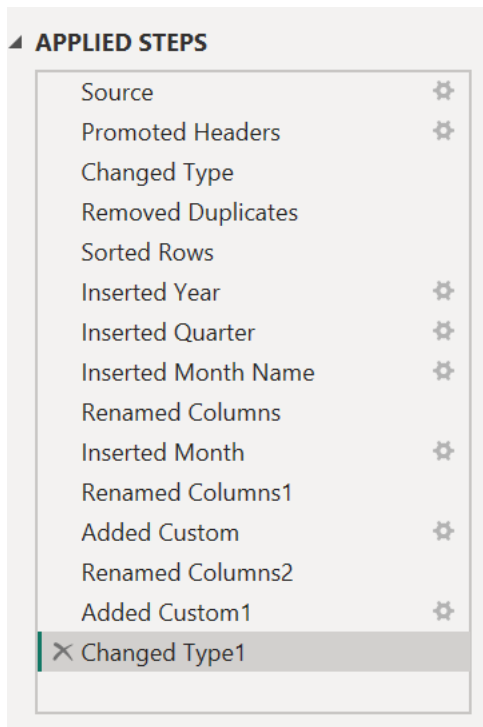
Data Transformation using Power Query

After importing the cleaned dataset into Power BI, additional transformations were performed using Power Query Editor to prepare the data for analysis.

Power Query provides a visual interface for transforming and shaping data before loading it into the data model.

Applied Steps

Step	Purpose
Source	Connected to the cleaned CSV dataset.
Promoted Headers	Converted the first row into column headers.
Changed Type	Assigned correct data types to all columns.
Removed Duplicates	Eliminated duplicate records to improve data integrity.
Sorted Rows	Sorted the dataset by Order Date in ascending order.
Inserted Year	Created a Year column from Order Date.
Inserted Quarter	Created a Quarter column for quarterly analysis.
Inserted Month Name	Generated Month Name for readable reporting.
Inserted Month Number	Created Month Number to ensure correct chronological sorting.
Added Custom Column – Age Group	Categorized customers into age groups for demographic analysis.
Added Custom Column – Discount Category	Grouped discounts into Low, Medium, and High categories.
Loaded Data	Loaded the transformed dataset into the Power BI data model.



These transformations improve the usability of the dataset by:

- Making time-based analysis easier.
- Enabling customer segmentation.
- Supporting discount analysis.
- Ensuring correct sorting of months.
- Improving dashboard performance and readability.

Derived Columns

Additional columns were created to enhance business analysis.

Age Group

Customers were grouped into meaningful age categories to understand purchasing behavior across different demographics.

Customer Age	Age Group
18–25	Young Adults
26–35	Adults
36–45	Middle Age
46+	Senior Adults

Business Benefit

This categorization helps identify which customer age groups contribute the most revenue and enables targeted marketing strategies.

Discount Category

The Discount column was categorized into three groups.

Discount (%) Category

0–15	Low Discount
15–30	Medium Discount
Above 30	High Discount

Business Benefit

Grouping discount percentages helps evaluate promotional strategies and determine whether higher discounts lead to increased sales.

Data Modelling

Data modelling is one of the most important stages in Power BI because it defines how tables interact and enables accurate calculations.

This project follows a **Star Schema** design.

Fact Table

Sales_Data

The fact table stores transactional sales records.

Key fields include:

- Order Date
 - Product Category
 - Sales Amount
 - Discount
 - Customer Age
 - Customer Gender
 - Sales Region
 - Sales Representative
-

Dimension Table

Calendar

A separate Calendar table was created to support time intelligence calculations and consistent date analysis.

The Calendar table contains:

- Date
 - Year
 - Quarter
 - Month
 - Month Number
-

Relationship

A **One-to-Many (1:*)** relationship was established:

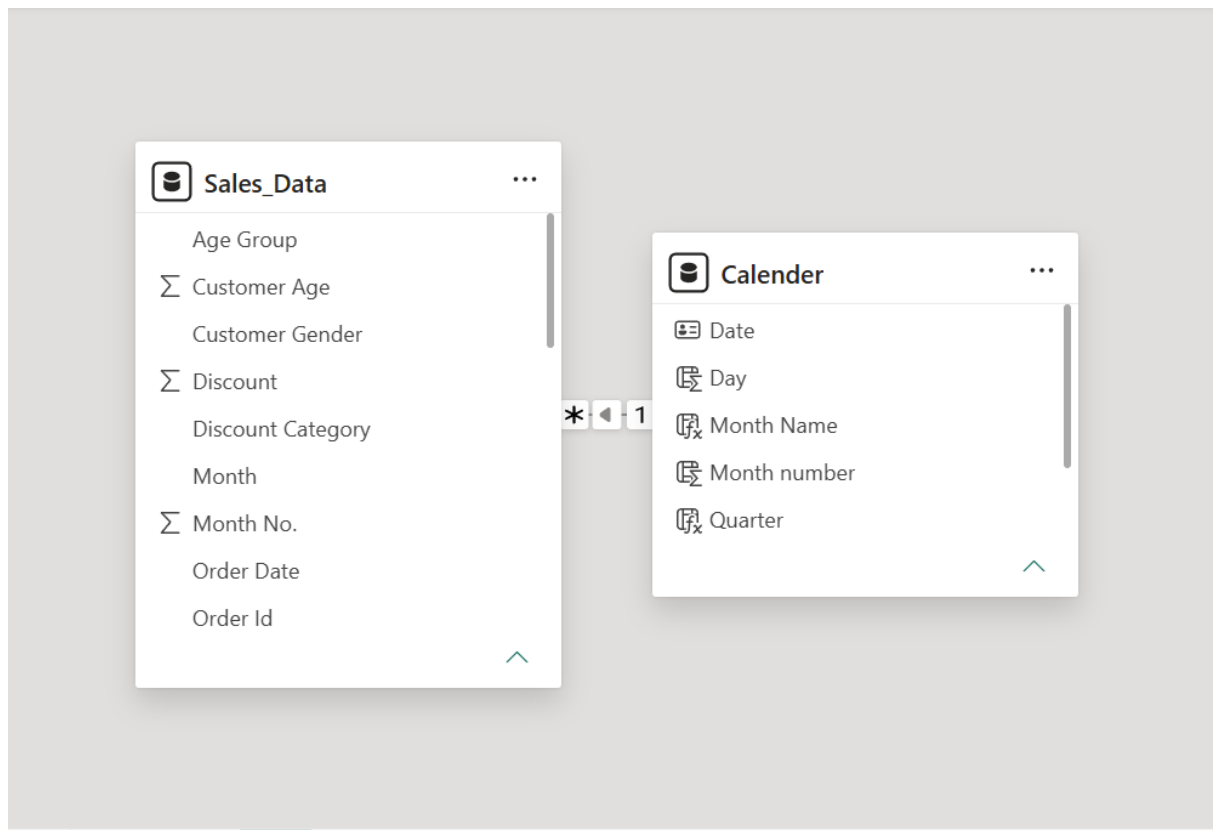
Calendar[Date]

|
| 1
|
|
| *

Sales_Data[Order Date]

Relationship Type:

- One-to-Many (1:*)
- Single Direction Filter



Why use a Calendar Table?

Creating a dedicated Calendar table provides several benefits:

- Supports time intelligence functions.
- Enables Year-to-Date (YTD) analysis.
- Improves filtering across reports.
- Ensures months appear in chronological order.
- Simplifies date-based calculations.

Data Validation

After completing the transformation and modeling process, the dataset was validated to ensure its accuracy.

The following checks were performed:

- Verified row count after cleaning.
- Confirmed correct data types.
- Checked for duplicate records.
- Validated relationships between tables.
- Ensured Month Name was sorted using Month Number.
- Verified DAX measures produced expected results.
- Cross-checked dashboard totals with the source data.

Validation Outcome

The dataset was successfully validated and found to be consistent, complete, and suitable for business analysis.

DAX Measures

Data Analysis Expressions (DAX) is the formula language used in Microsoft Power BI to create custom calculations and measures. DAX enables analysts to perform aggregations, filtering, time-based analysis, and complex business calculations that go beyond the capabilities of standard visualizations.

For this project, several DAX measures were created to calculate key performance indicators (KPIs) used throughout the dashboard.

1. Total Sales

Total Sales =
SUM(Sales_Data[Sales Amount])

Purpose

Calculates the total revenue generated from all completed sales transactions.

Business Importance

- Measures overall business performance.
- Used by management to monitor revenue.
- Forms the basis for many other KPIs.

Dashboard Visual

KPI Card

2. Total Transactions

Total Transactions =
COUNTROWS(Sales_Data)

Purpose

Counts the total number of sales transactions.

Business Importance

- Indicates business activity.
- Helps evaluate sales volume.

Dashboard Visual

KPI Card

3. Average Sales

Average Sales =

AVERAGE(Sales_Data[Sales Amount])

Purpose

Calculates the average value of each sales transaction.

Business Importance

- Helps evaluate average customer spending.
- Useful for pricing and promotional strategies.

Dashboard Visual

KPI Card

4. Average Discount

Average Discount =

AVERAGE(Sales_Data[Discount])

Purpose

Calculates the average discount percentage provided to customers.

Business Importance

- Monitors promotional effectiveness.
 - Supports pricing decisions.
-

5. Average Customer Age

Average Customer Age =

AVERAGE(Sales_Data[Customer Age])

Purpose

Calculates the average age of customers.

Business Importance

- Helps understand the primary customer demographic.
 - Supports targeted marketing campaigns.
-

Additional Analytical Measures

If you created any of these, include them in your report.

Sales by Age Group

Sales 26-35 =

```
CALCULATE(  
    [Total Sales],  
    Sales_Data[Age Group] = "26-35"  
)
```

Female Sales with High Discount

Female Sales High Discount =

```
CALCULATE(  
    [Total Sales],  
    FILTER(  
        Sales_Data,  
        Sales_Data[Customer Gender]="F"  
        &&  
        Sales_Data[Discount]>30  
    )  
)
```

Why DAX?

DAX allows analysts to create reusable business calculations that automatically respond to filters and slicers, enabling interactive dashboards and real-time analysis.

Dashboard Design Approach

The dashboard was designed to provide management with a clear, interactive, and user-friendly view of retail sales performance.

The design focused on:

- Simplicity
- Readability
- Business storytelling
- Interactive analysis
- Quick decision-making

The dashboard layout follows a **top-to-bottom information hierarchy**.

Dashboard Layout

KPI Cards



Sales Trend



Product Analysis



Regional Analysis



Customer Analysis



Sales Representative Analysis



Filters / Slicers

Color Selection

A consistent blue color theme was selected to create a professional appearance while maintaining readability.

User Experience

Interactive slicers allow users to analyze data based on:

- Product Category
- Sales Region
- Customer Gender
- Age Group
- Month

This enables users to explore the dashboard without modifying the underlying dataset.

Dashboard Overview



KPI Cards

The KPI cards provide a quick summary of the organization's overall performance.

Displayed KPIs include:

- Total Sales
- Total Transactions
- Average Sales
- Average Discount
- Average Customer Age

These KPIs give decision-makers an immediate understanding of business performance.

Monthly Sales Trend

A line chart displays sales trends across months.

Business Purpose:

- Monitor monthly performance.
- Identify seasonal fluctuations.
- Detect unusual increases or decreases in sales.

Product Category Analysis

A bar chart compares revenue across different product categories.

Business Purpose:

- Identify top-performing categories.
 - Support inventory planning.
 - Evaluate category profitability.
-

Regional Sales Analysis

A map visual displays sales across different geographic regions.

Business Purpose:

- Compare regional performance.
 - Identify high-performing and underperforming regions.
 - Support regional expansion decisions.
-

Customer Age Analysis

A bar chart shows revenue by age group.

Business Purpose:

- Identify the most valuable customer segment.
 - Support targeted marketing campaigns.
-

Customer Gender Analysis

A donut chart displays the proportion of sales by customer gender.

Business Purpose:

- Compare purchasing behavior between customer groups.
 - Support demographic analysis.
-

Sales Representative Performance

A bar chart ranks sales representatives based on total sales.

Business Purpose:

- Identify top performers.
- Support performance reviews.
- Encourage knowledge sharig within the sales team.

Why These Visuals Were Chosen

Visual	Purpose	Business Benefit
KPI Cards	Display summary metrics	Immediate performance overview
Line Chart	Monthly trend analysis	Detect seasonal changes
Bar Chart	Product comparison	Identify top categories
Map	Geographic analysis	Compare regional sales
Donut Chart	Gender distribution	Understand customer demographics
Bar Chart	Sales representative ranking	Evaluate employee performance
Slicers	Interactive filtering	Customized analysis

Dashboard Design Principles

The dashboard follows best practices in data visualization:

- Minimal clutter
- Consistent formatting
- Logical layout
- Clear labels
- Appropriate chart selection
- Interactive filtering
-

These principles improve readability and support effective decision-making.

Testing & Validation

After developing the dashboard, several validation checks were performed to ensure accuracy and reliability.

Validation Checklist

- KPI values matched the source dataset.
 - DAX measures returned expected results.
 - Filters updated all visuals correctly.
 - Month names displayed in chronological order.
 - Interactive slicers worked across visuals.
 - Relationships produced accurate calculations.
 - Dashboard performance remained responsive.
-

Project Outcome

The completed dashboard provides stakeholders with a centralized platform for monitoring sales performance, understanding customer behavior, evaluating regional trends, and supporting business decisions.

The use of Power BI, Power Query, and DAX transformed raw transactional data into meaningful business insights through interactive visualizations.

Business Insights

The Power BI dashboard provides a comprehensive overview of retail sales performance by combining key business metrics with interactive visualizations. Based on the analysis of the cleaned dataset, several important business insights were identified.

Overall Sales Performance

The organization generated **\$49.52 Million** in total sales from **90,000** completed transactions between **January 2024 and September 2024**.

The average sales value per transaction was **\$550.17**, indicating a healthy average purchase amount.

Business Interpretation

- The business maintained strong revenue throughout the analysis period.
 - The average transaction value suggests customers are purchasing multiple products or higher-value items.
 - The KPIs provide management with a quick snapshot of overall business performance.
-

Monthly Sales Trend

The monthly sales trend remained relatively stable throughout most of the analysis period. Sales were consistent from **January through August**, while **September recorded comparatively lower sales**.

Since the dataset contains data only until September 2024, no conclusions are drawn beyond the available period.

Business Interpretation

- Stable sales indicate consistent customer demand.
 - The lower sales in September should be investigated to determine whether the cause is seasonal demand, operational issues, or incomplete monthly data.
-

Customer Demographics

Customer segmentation based on age revealed that the **46+ age group generated the highest revenue** among all customer segments.

Sales distribution between male and female customers was relatively balanced, suggesting that both customer groups contribute significantly to overall revenue.

Business Interpretation

- The 46+ customer segment represents the most valuable customer base.
 - Marketing campaigns targeting this segment could improve customer retention and repeat purchases.
 - Balanced gender distribution indicates broad market appeal.
-

Product Category Performance

Revenue was distributed across multiple product categories rather than being dominated by a single category.

Business Interpretation

- A diversified product portfolio reduces dependency on one product line.
 - Lower-performing categories should be monitored for pricing, inventory, or promotional improvements.
 - High-performing categories should continue receiving marketing and inventory support.
-

Regional Sales Performance

Sales were generated across multiple geographic regions, demonstrating broad market coverage.

Business Interpretation

- High-performing regions can serve as benchmarks for sales strategies.
 - Underperforming regions should be reviewed for opportunities to improve market penetration, promotions, or product availability.
-

Sales Representative Performance

Sales representatives showed varying levels of performance, with the top performers generating approximately \$17K–19K in sales.

Business Interpretation

- Top-performing representatives can mentor other team members.
- Performance analysis can support recognition programs and targeted training initiatives.

Business Recommendations

Based on the analysis performed, the following recommendations are proposed to improve business performance.

Recommendation 1 – Strengthen Customer Retention

Since customers aged **46 years and above** generated the highest revenue, the organization should introduce targeted loyalty programs, personalized offers, and retention campaigns for this valuable customer segment.

Expected Benefit

- Increased customer loyalty
 - Higher repeat purchases
 - Improved customer lifetime value
-

Recommendation 2 – Improve Sales Team Performance

Analyze the sales strategies used by the highest-performing sales representatives and incorporate these practices into training programs for the entire sales team.

Expected Benefit

- Improved productivity
 - Better sales consistency
 - Increased overall revenue
-

Recommendation 3 – Monitor Regional Performance

Conduct monthly reviews of regional sales performance to identify underperforming markets and allocate marketing resources effectively.

Expected Benefit

- Better regional growth
 - Improved resource allocation
 - Increased market penetration
-

Recommendation 4 – Optimize Product Strategy

Review product categories with relatively lower revenue and evaluate opportunities for promotional campaigns, pricing adjustments, or inventory optimization.

Expected Benefit

- Improved category performance
 - Better inventory utilization
 - Increased profitability
-

Recommendation 5 – Monitor Sales Trends

Regularly analyze monthly sales trends to identify unexpected fluctuations and investigate the underlying causes.

Expected Benefit

- Early identification of business risks
 - Better forecasting
 - Improved strategic planning
-

Recommendation 6 – Expand Dashboard Capabilities

Enhance the dashboard in future versions by incorporating:

- Year-over-Year Analysis
- Sales Forecasting
- Drill-through Reports
- Row-Level Security (RLS)
- Power BI Service publishing
- Automated data refresh

Expected Benefit

- Enhanced decision-making
 - Improved reporting capabilities
- Better scalability

Challenges Encountered During the Project

Challenge 1 – Missing Values

Problem

The original dataset contained **10,000 records** with missing values in key fields, including Sales Amount, Discount, Customer Age, and Customer Gender.

Solution

The incomplete records were removed in Microsoft Excel before importing the dataset into Power BI, resulting in a clean dataset of **90,000 records**.

Challenge 2 – Month Sorting

Problem

Month names were displayed alphabetically rather than chronologically.

Solution

A Month Number column was created and used to sort the Month Name column correctly.

Challenge 3 – Customer Segmentation

Problem

Customer age values were difficult to interpret directly.

Solution

A custom **Age Group** column was created to classify customers into meaningful demographic categories.

Challenge 4 – Discount Analysis

Problem

Individual discount percentages were difficult to analyze.

Solution

A **Discount Category** column was created to group discounts into Low, Medium, and High ranges.

Challenge 5 – Time-Based Analysis

Problem

The dataset contained only one year of sales data.

Solution

A dedicated Calendar table was created to enable proper time intelligence and future scalability.

Technical Skills Demonstrated

Business Intelligence

- Microsoft Power BI
- Interactive Dashboard Development
- KPI Reporting

Data Preparation

- Microsoft Excel
- Data Cleaning
- Missing Value Handling
- Duplicate Removal

Power Query

- Data Transformation
- Custom Columns
- Data Type Validation

Data Modeling

- Star Schema
- Calendar Table
- Relationship Management

DAX

- Aggregation Functions
- CALCULATE()
- FILTER()
- Time Intelligence
- Business KPIs

Data Visualization

- KPI Cards
- Bar Charts
- Line Charts
- Maps
- Donut Charts
- Interactive Slicers

Business Analysis

- Sales Analysis
- Customer Segmentation
- Regional Performance Analysis
- Product Performance Evaluation
- Business Recommendations

Future Enhancements

This dashboard can be further enhanced by implementing the following features:

- Multi-year sales analysis
 - Year-over-Year (YoY) growth metrics
 - Sales forecasting using Power BI forecasting tools
 - Drill-through reports for detailed analysis
 - Bookmarks and navigation buttons
 - Role-Level Security (RLS)
 - Power BI Service deployment with scheduled refresh
 - Mobile-optimized dashboard layout
 - Integration with SQL Server or Azure Data Factory
 - AI-powered visuals such as Key Influencers and Decomposition Tree
-

Conclusion

The **Retail Sales Performance Dashboard** project demonstrates the complete business intelligence lifecycle, beginning with data preparation and ending with actionable business insights.

Using Microsoft Excel, Power Query, DAX, and Power BI, raw transactional data was transformed into an interactive dashboard that enables users to monitor sales performance, evaluate customer behavior, analyze regional trends, and support data-driven decision-making.

The project strengthened my skills in:

- Data Cleaning
- Data Transformation
- Data Modeling
- DAX Development
- Dashboard Design
- Business Analysis
- Data Storytelling

This project also highlights my ability to convert raw data into meaningful insights that support strategic business decisions. It serves as a strong addition to my professional portfolio and demonstrates practical, end-to-end Power BI skills.

References

1. Kaggle – Retail Sales Dataset
 2. Microsoft Learn – Power BI Documentation
 3. Microsoft Learn – DAX Documentation
 4. Microsoft Learn – Power Query Documentation
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